

Fashion-MNIST Image Classification Using Convolutional Neural Networks

Professional Project Documentation

Submitted By	Shafaat Ullah
Project Type	Deep Learning / Computer Vision
Model	Convolutional Neural Network (CNN)
Dataset	Fashion-MNIST
Input Image	28 × 28 Grayscale
Number of Classes	10
Framework	TensorFlow / Keras

This document presents the complete workflow of a Fashion-MNIST image-classification project using a Convolutional Neural Network, including data preparation, CNN architecture, training, evaluation, error analysis, and troubleshooting.

1. Introduction

Image classification is an important application of computer vision and deep learning. This project develops a Convolutional Neural Network (CNN) to automatically recognize different categories of fashion products from grayscale images.

2. Problem Statement

The objective is to classify each 28×28 grayscale fashion image into one of ten predefined categories. CNNs are suitable because they can automatically learn spatial and visual features directly from image data.

3. Project Objectives

The main objectives are to prepare the Fashion-MNIST dataset, normalize and reshape the images, design a CNN model, train the model using labeled images, evaluate predictions on unseen test data, and identify common classification errors.

4. Dataset Description

Fashion-MNIST contains grayscale images of fashion products. Each image has a resolution of 28×28 pixels. The ten classes are T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot.

5. Data Preprocessing

Pixel values are converted to floating-point values and normalized from the 0–255 range to 0–1. The images are reshaped to include a single grayscale channel, producing an input shape of $28 \times 28 \times 1$ for the CNN.

6. CNN Architecture

The model follows the standard CNN workflow: convolution layers extract visual features, ReLU introduces non-linearity, pooling reduces spatial dimensions, Flatten converts feature maps into a vector, Dense layers perform high-level classification, and Softmax produces probabilities for the ten classes.

7. Convolution Layer

Convolution applies learnable filters to local regions of an image. As a filter moves across the image, it detects patterns and creates a feature map. Early filters can learn edges and simple shapes, while deeper layers can represent more complex visual structures.

8. ReLU Activation

The Rectified Linear Unit (ReLU) activation function replaces negative values with zero while preserving positive values. It introduces non-linearity and allows the network to learn complex relationships in image data.

9. Pooling Layer

Max Pooling reduces the spatial dimensions of feature maps while preserving strong detected features. This decreases computational cost and provides some robustness to small changes in feature location.

10. Flatten and Dense Layers

After convolution and pooling, the resulting feature maps are flattened into a one-dimensional vector. Dense layers use these learned features to determine the most appropriate fashion category. Dropout can help reduce overfitting.

11. Output Layer

The output layer contains ten units corresponding to the ten Fashion-MNIST classes. Softmax converts the output into class probabilities. The class with the highest probability becomes the predicted label.

12. Model Training

The CNN is trained using labeled training images. During training, the model calculates classification loss, updates its learnable parameters, and gradually improves its predictions. The Adam optimizer is used for gradient-based optimization.

13. Errors Encountered and Troubleshooting

During development, common notebook errors were encountered, including variables such as NumPy, labels, or the model not being defined before use. These issues were resolved by executing cells in the correct order and ensuring that all required imports, datasets, labels, and model definitions were available before prediction or evaluation. A Keras input-shape warning and a CUDA initialization message were also observed. The input-shape issue can be handled by defining the model input explicitly, while the CUDA message generally indicates that GPU acceleration is unavailable and does not prevent CPU-based model execution.

14. Model Evaluation

The trained model is evaluated on unseen test images. Overall accuracy measures the proportion of correctly classified images. Class-specific error analysis is also useful because some clothing categories have visually similar characteristics.

15. Pullover vs Shirt Error Analysis

A focused analysis was performed for confusion between Pullover and Shirt. The recorded results showed 80 Pullover images classified as Shirt and 37 Shirt images classified as Pullover, resulting in 117 mistakes between these two categories.

16. Results

The CNN successfully completed the Fashion-MNIST classification workflow and produced predictions for the test dataset. The Pullover–Shirt analysis identified a specific source of confusion and provides useful information for future model refinement.

17. Limitations

Fashion-MNIST images are small and grayscale, so some visually similar clothing categories are difficult to distinguish. Similar silhouettes and textures can lead to confusion between related classes.

18. Future Improvements

Possible improvements include data augmentation, batch normalization, learning-rate scheduling, hyperparameter tuning, regularization, deeper architectures, and detailed confusion-matrix analysis.

19. Conclusion

This project demonstrates a complete practical deep-learning workflow for image classification. The CNN learns visual features from Fashion-MNIST images and uses them to classify ten clothing categories. The project also demonstrates the value of analyzing specific errors rather than relying only on overall accuracy.

Project Summary

Component	Description
Submitted By	Shafaat Ullah
Dataset	Fashion-MNIST
Input	28 × 28 grayscale images
Task	10-class image classification
Core Pipeline	Convolution → ReLU → Pooling → Flatten → Dense → Softmax
Evaluation	Test-set prediction and class-specific error analysis
Focused Analysis	Pullover ↔ Shirt confusion
Recorded Errors	80 + 37 = 117